

ELECTRIC VEHICLE TECHNOLOGY INTERNSHIP

TASK 1: EV SYSTEM STUDY

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Objective: To study the major components of an electric vehicle, understand their functions and interaction, and present the EV system through a technical block diagram.

1. Introduction

An Electric Vehicle (EV) uses electrical energy stored in a rechargeable battery to produce traction through an electric motor. Unlike conventional vehicles that primarily depend on an internal combustion engine, EVs integrate electrochemical energy storage, power electronics, electric machines, control systems and charging infrastructure. The major systems operate together to deliver controlled torque to the wheels while maintaining safe and efficient energy flow.

2. Major Components of an Electric Vehicle

Component	Primary function
Battery Pack	Stores electrical energy as DC energy and supplies power to the traction system.
Battery Management System (BMS)	Monitors cell voltage, current and temperature; estimates battery state and provides protection.
Electric Motor	Converts electrical energy into mechanical energy for vehicle propulsion.
Inverter	Converts and controls DC power from the battery into the required AC power for many traction motors.
Vehicle / Motor Controller	Processes driver commands and sensor information to control torque, speed and operating mode.
On-board Charger (OBC)	Converts AC charging power into regulated DC power suitable for charging the battery.
DC-DC Converter	Converts high-voltage battery power to lower voltage for auxiliary electrical systems.
Charging Port	Provides the electrical interface between the EV and an external charging supply.
Reduction Gear & Differential	Matches motor speed and torque to wheel requirements and distributes mechanical power.
Regenerative Braking System	Recovers part of the vehicle's kinetic energy during braking and sends electrical energy back to the battery.

3. Roles of Key EV Components

Battery: The battery pack is the primary energy-storage unit. It supplies DC electrical energy to the high-voltage traction system. Battery capacity is commonly expressed in kWh, while voltage and current determine the electrical power available to the system.

Electric Motor: The traction motor converts electrical power into mechanical power. Depending on vehicle design, motors such as PMSM, BLDC or induction motors may be used. Motor torque is responsible for accelerating the vehicle.

Inverter: The inverter is a major power-electronic converter between the high-voltage battery and the traction motor. In a typical AC traction system it converts DC from the battery into controlled AC waveforms and regulates motor operation.

Controller: The vehicle or motor controller receives inputs such as accelerator position, braking demand and sensor feedback. It determines the required torque and sends appropriate control commands to the power electronics and motor.

Charger: During AC charging, the on-board charger converts incoming AC power into controlled DC power for the battery. The charging system also communicates with the vehicle and follows charging limits defined by the battery and BMS.

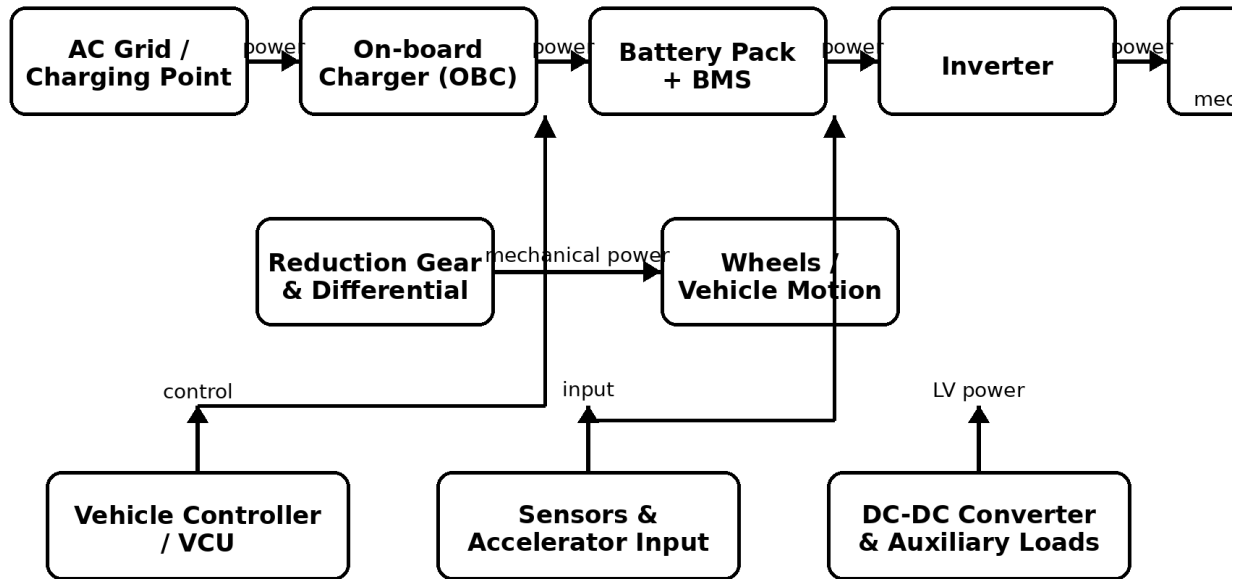
BMS: The Battery Management System monitors important battery parameters including cell voltage, pack current and temperature. It helps maintain safe operation and can balance cells to improve pack performance and usable life.

4. Interaction of Major Systems

During propulsion, electrical energy flows from the battery to the inverter and then to the electric motor. The controller determines the required motor torque from driver inputs and sensor feedback. The motor converts the controlled electrical power into mechanical power, which is transferred through the reduction gear and differential to the wheels. During charging, electrical energy flows from the grid through the charging interface and charger into the battery. During regenerative braking, the energy flow can partially reverse: the rotating motor acts as a generator and electrical energy is returned to the battery.

5. EV System Block Diagram

Electric Vehicle (EV) System Block Diagram



Regenerative braking: motor can operate as a generator and return electrical energy to the battery.

6. Advantages of Electric Vehicles

- High energy efficiency compared with conventional internal-combustion propulsion.
- Zero tailpipe emissions during vehicle operation.
- Regenerative braking can recover part of the vehicle's kinetic energy.
- Fewer moving parts in the traction drivetrain can reduce mechanical complexity.
- Electric motors can provide high torque from low speed.
- EVs can be integrated with renewable-energy-based charging systems.

7. Technical Observations

The overall performance of an EV depends on the coordinated operation of energy storage, power electronics, motor control, thermal management and charging systems. Battery voltage and capacity affect the available energy and power, while inverter and motor efficiency influence how effectively stored energy is converted into vehicle motion. Safe operation requires continuous monitoring by the BMS and appropriate control of current, voltage and temperature.

8. Conclusion

An electric vehicle is an integrated electro-mechanical system in which the battery, BMS, inverter, controller, motor, charger and drivetrain perform complementary functions. Understanding the energy flow and interaction among these components provides the foundation for further study of EV batteries, charging systems, power electronics and electric machines. This EV System Study establishes the basic technical knowledge required for deeper analysis of EV battery and charging technologies.

9. References

1. General principles of electric vehicles, electric machines and power electronics from standard engineering textbooks and manufacturer technical documentation.
2. Battery Management System and EV charging concepts from standard academic references.
3. SkillZeno Electric Vehicle Technology Internship — Task 1 requirements.